**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 060 · Combining Programming Constructs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why must a trace table follow executed assignments in order?

Q2. Which type suits a count of pupils?
A. Real
B. Boolean
C. Character
D. Integer

Q3. How do character and string data differ?

Q4. A limit of 30 appears five times in code. One copy changes to 32; the rest are missed. How would a named constant help? How does it differ from a changing score?

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# I Do · Read new knowledge

## Key vocabulary

Sequence — Initialise variables and read inputs before using their values.

Selection — Choose a route after evaluating a Boolean condition.

Iteration — Repeat a block using a loop where a task demands it.

Combined constructs — Real programs often need sequential code, decisions and repetition together.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

## New learning

Sequence: Initialise variables and read inputs before using their values. For example, consider a number guessing game.

Selection: Choose a route after evaluating a Boolean condition. For example, consider a quiz scorer.

Iteration: Repeat a block using a loop where a task demands it. For example, consider a book reminder.

Combined constructs: Real programs often need sequential code, decisions and repetition together. For example, consider a class rewards.

## Worked example

A number guessing game is being developed. Explain how sequence and selection are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Sequence [1]; Initialise variables and read inputs before using their values. [1].

3. Selection [1]; Choose a route after evaluating a Boolean condition. [1].

4. Give one accurate point and a linked reason for each technique.

# We Do · Practise with guidance

### Guided 1

A quiz scorer is being programmed. Explain why sequence matters. [2 marks]

Worked method: Sequence [1]; Initialise variables and read inputs before using their values. [1].

### Guided 2

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For a book reminder, explain how selection is used. [2 marks]

Worked method: Selection [1]; Choose a route after evaluating a Boolean condition. [1].

**Guided 3**

A class rewards needs iteration and combined constructs. Explain the roles of both. [4 marks]

Worked method: Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1].

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the condition to print only the even device IDs.

```
for device = 1 to 4
    if ... then
        print(device)
    endif
next device
```

write

### Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

### Question 2

Explain how selection and iteration can help a repeat ordering. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

### Question 3

Explain the roles of sequence, iteration and combined constructs when creating a payment kiosk. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

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## Exit Ticket · Answer before checking

Variables should be initialised and required inputs read before their values are used.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Each variable value depends on the state produced by earlier executed instructions.

Q2. D — Integer

Q3. A character represents one character; a string represents a sequence of characters.

Q4. Define one named constant, such as MAX_CLASS_SIZE, and use it throughout. Change its definition once to update the limit. Use a variable for a score because it changes while the program runs.

## I Do / We Do pages · Guided practice

Q1. Sequence [1]; Initialise variables and read inputs before using their values. [1].

Q2. Selection [1]; Choose a route after evaluating a Boolean condition. [1].

Q3. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1].

## You Do pages · Code First

```
for device = 1 to 4
    if device MOD 2 == 0 then
        print(device)
    endif
next device
```

Expected result: 2 4

## You Do pages · Independent questions

Q1. Sequence [1]; Iteration [1].

Q2. Selection [1]; Choose a route after evaluating a Boolean condition. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. Sequence [1]; Initialise variables and read inputs before using their values. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**